

Failure risk assessment of interdependent infrastructures against earthquake, a Petri net approach: case study—power and water distribution networks

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Abstract Metropolitan areas consist of complicated systems of interconnected infrastructures that are highly interdependent. Disruption of one infrastructure may induce disruption in other interconnected ones. The results from analysis of one infrastructure as an independent system are not realistic without considering the behavior of other interconnected infrastructures. Consequently, the study of the interdependencies among critical infrastructures is important for addressing the cascading effects of a failed infrastructure on the entire network to properly model its performance and help the disaster management team in decision making. In this study, the extended Petri net and Markov chain have been used to demonstrate the power and water infrastructure interdependency with a case study of one of the municipal districts of metropolitan Tehran, the capital of Iran. In this research, three cases have been assessed quantitatively: (1) the intra-dependency effects of different components in each network, (2) the interdependency effects between the considered critical infrastructures and (3) the behavior of the water network considering intra- and interdependency, when the power network fails. The analyses show that considering the mentioned interdependencies has a major influence on their performance simulations and assessment of their exact vulnerability. It is concluded that the failure probability of the water network that is dependent on the failed power network is 1.66 of the independent water network in the studied region. Eventually, the results of the research could be used in design, restoration and disaster management planning for safety assessment of critical infrastructures.

Keywords Critical infrastructures interaction · Petri net · Markov chain · Water network · Power network · Network analysis · Tehran

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1 Introduction

Realistic vulnerability assessment and seismic fragility estimation are key elements in the process of decision making for mitigating earthquake risks to intra- and interdependent infrastructures. When the fragility of a component or facility in a networked infrastructure, the vulnerability of an infrastructure system as a whole or the risk of a set of interconnected infrastructures as a system of systems are considered, the performance of a facility in the network may be affected by the functionality of other components, facilities and inter-connected systems.

Therefore, there is a need for modeling and simulation of interdependent infrastructures in performance assessment and risk assessment and management. Interdependencies could be divided into four categories: physical, geographical, informational and logical. Physical interdependencies are encountered when the state of the infrastructures depends on tangible or material linkages among them. The quantification of physical interdependency has low uncertainty. Visual inspection is the simplest way for determining physical coupling. Dependency based on spatial proximity among network elements is also quantifiable as geographical interdependency. Informational interdependencies occur due to the contemporary need for exchange of data and the computerized control of infrastructure behavior. Finally, logical interdependencies emerge when human decisions intervene or when the relation is not physical, geographical or informational (Rinaldi et al. 2001).

In order to perform a realistic study on the behavior and operation of infrastructures, the modeling and simulation of interdependencies is essential. The study of interdependent infrastructures is in the early stages, and it is placed in a challenging category of study. A rather complete review and conceptualization of available research in the field of infrastructure interdependencies have been done by Satumtira and Dueñas-Orsorio (2010). They have integrated the multiple ideas in this area into a unified hierarchical structure that navigates through research advances from early papers in the 1980s to the present time.

Petri net is a network modeling tool based on graph theory. It is very robust for modeling, simulation and analysis of the network behavior. It was originally introduced by Carl Adam Petri in the early 1960s (Petri 1962). Petri nets have been used to model various kinds of dynamic event-driven systems. It was improved and incorporated later by some researchers (Peterson 1981; Murata 1989; Bobbio 1990). Two categories of Petri net, namely basic Petri net and extended Petri net, have been used by researchers for modeling infrastructures. The basic Petri net can be used to determine the interdependencies among the infrastructure network. Gursesli and Desrochers (2003) applied the graph-based Petri net to identify the interdependencies among critical urban infrastructures defined by Rinaldi et al. (2001). The modeled network consisted of electric power, oil, transportation, natural gas, telecommunications and water sectors as critical infrastructure. The basic Petri net model was analyzed to examine the interdependencies among the infrastructures due to the failure of power plant as the main supporting infrastructure, and it was shown that the Petri net is a very strong modeling tool to capture the interrelationships among the infrastructures.

The basic Petri net does not consider the firing time of the transitions. To account for the required time of the occurrence of an event, it is assumed that the firing time of each transition is an exponentially distributed random variable. Thus, the extended Petri net is called the stochastic Petri net. Ramchandani (1974) was the first who introduced techniques of executing extended Petri net as the extension of original Petri net theory to a stochastic case. Later, Zuberek studies (1991) made the methodology clearer for practical application of the theory in real-life problems. Zuberek (1991) has defined the

characteristics of the timed Petri net. The basic concept is that in the timed Petri net, each transition takes a positive time to fire and the firing time is an exponential random variable. Cirado and Lindemann (1993) introduced a time- and space-efficient algorithm for achieving steady-state solutions of deterministic and stochastic Petri net (DSPN) with both stochastic and structural analyses. The reachability graph of SPN can be compared with the inherent Markov chain, which can be solved analytically to determine the probabilities of the states of network (Murata 1989; Bobbio 1990).

Krings and Oman (2003) introduced a simple generalized stochastic Petri net model for modeling the cascading failures of critical infrastructures. Immediate and timed events were included in their studies for identifying common-mode faults. Sultana and Chen (2009) modeled flood-related interdependencies among hydroelectricity generating infrastructures using extended Petri net and Markov chain analysis. They calculated the relative probability of each set of events resulted in total failure of the system known as absorbing state. The relative probability of each failure state is calculated and compared with the other failure states. The relative frequency of failure states is calculated considering the total set of failure states. Luna et al. (2011) proposed an improved model for the disaster restoration process by using a discrete event simulation approach based on colored Petri nets. They modeled the trunk network of the Tokyo water distribution system and simulated its behavior in case of a rare event such as an earthquake. The timed simulation allowed visualizing the restoration progress, which is usually depicted as a restoration curve.

In addition to the fact that functions of components within an infrastructure system are intra-dependent on each other, the functions of infrastructure systems are also interdependent. Rinaldi et al. (2001), Peerenboom et al. (2002), Gillette et al. (2002), NERC (2004) and Fedora (2004) noted the importance of infrastructure interdependencies on economic, industrial and social activities. Rinaldi et al. (2001) and Peerenboom et al. (2002) classified infrastructure interdependencies based on relationships among infrastructure systems. Chang et al. (1996) also noted the potential importance of infrastructure interdependency among the water distribution, the gas distribution and the electrical power transmission systems in Shelby County, TN, but did not determine the quantitative effects of infrastructure interdependency on serviceability. The study by Robert (2004) is one of the first that attempted to assess network serviceability considering system interdependency effects directly. Interdependency among systems was modeled by direct links and indirect links. A direct link represents physical attachment of two facilities. Though Robert's study recognizes the existence of infrastructure interaction effects and their importance when serviceability of a lifeline network is analyzed, it is applied to a hydroelectric power network, which is composed of a hydroelectric power generating station, a spillway, a dam, a storage basin and a river and a power distribution network, which is composed of electrical power substations and transmission lines. The details of the methodology used to analyze the effect of infrastructure interdependencies were not reported.

In the graph theory approach of Dueñas-Osario et al. (2007a, b), the probability that a node or an edge is removed is determined not only by the component fragility or the leakage or break ratio of a distribution element but also by the probability that a facility did not receive commodities which are needed to maintain the function of the nodes or the edges. Many studies (e.g., Chang et al. 1996; FEMA 1997, 2003) have considered financial losses incurred due to the loss of serviceability arising from system interdependency effects using input–output analysis (Haimes and Jiang 2001; Haimes et al. 2005a, b). Ge et al. introduced the Petri net into a geographical information system to develop the GeoPetri Net system, which was used to simulate the complex geographical relationships among places and nodes in interdependent infrastructures. The GeoPetri Net deals with

places and nodes with geographical locations and the geographical relationships between nodes and records the statuses of nodes to produce simulated events. On the basis of the simulated events, the posterior probability of the occurrence of each node can be obtained by calculating the number of coincidences of nodes and infrastructure networks to further analyze the interdependencies between nodes and infrastructure networks. A case study involving an education layer with 15 nodes (schools) and a transportation layer with 25 node lines (streets) in a geographical information system was presented to substantiate the proposed conceptual arguments (Ge et al. 2010).

In order to calculate the failure risk of interconnected systems, it is necessary to establish and consider different success and failure scenarios, and the absolute value of failure probabilities must be calculated. In other words, each sequence of events that results in the total failure has a unique occurrence probability. In this research, the absolute probabilities of failure scenarios are calculated considering event occurrence probabilities (known as fire rate of transition) as well as failure probability of components which are failed in that scenario. This proposed procedure is not applied in the available researches. In the present research, the failure risk of the interconnected water and electricity power infrastructures of the district is investigated quantitatively using extended Petri net and Markov chain theory. Four distinct cases on dependency, as independent facilities in each infrastructure, intra-dependent facilities in each infrastructure, interdependent infrastructures and conditional interdependent water infrastructure (electricity is completely out), are investigated and the necessity of this kind of modeling is deduced. In this research, the numerical results of the effect of interdependency and/or intra-dependency of systems and their components on seismic failure risk of two real interconnected infrastructure systems are presented. This is the first time that the failure probability of different intra-/interdependency cases, i.e., the intra-dependency effects of different components in each network, the interdependency effects between the considered critical infrastructures and the failure risk of the water network considering intra- and interdependency, when the power network fails, is compared in a quantitative manner in a real scale using the known powerful tool “Petri net.”

2 Methodology

The proposed algorithm for quantitative vulnerability assessment of interdependent infrastructures against earthquake is shown in Fig. 1. Modeling interconnected systems with Petri net and analyzing it with Markov chain is an efficient method in the evaluation of the vulnerability of interacting networks due to its ability in modeling the dependencies and the graphical relation between them and also their strong mathematical graph theory basis.

The steps of the proposed algorithm are described in this section.

2.1 Identification of systems

This step consists of three elements of identifying subsystems and components of considered networks, finding the geographical location of subsystems of the networks, and establishing interdependency and intra-dependency between subsystems of each network and components of different networks. In this study, intra-dependency is considered as functional dependency between components and subsystems of a network system. The interdependency is considered as the functional dependency between a component of one system and another component in the other network.

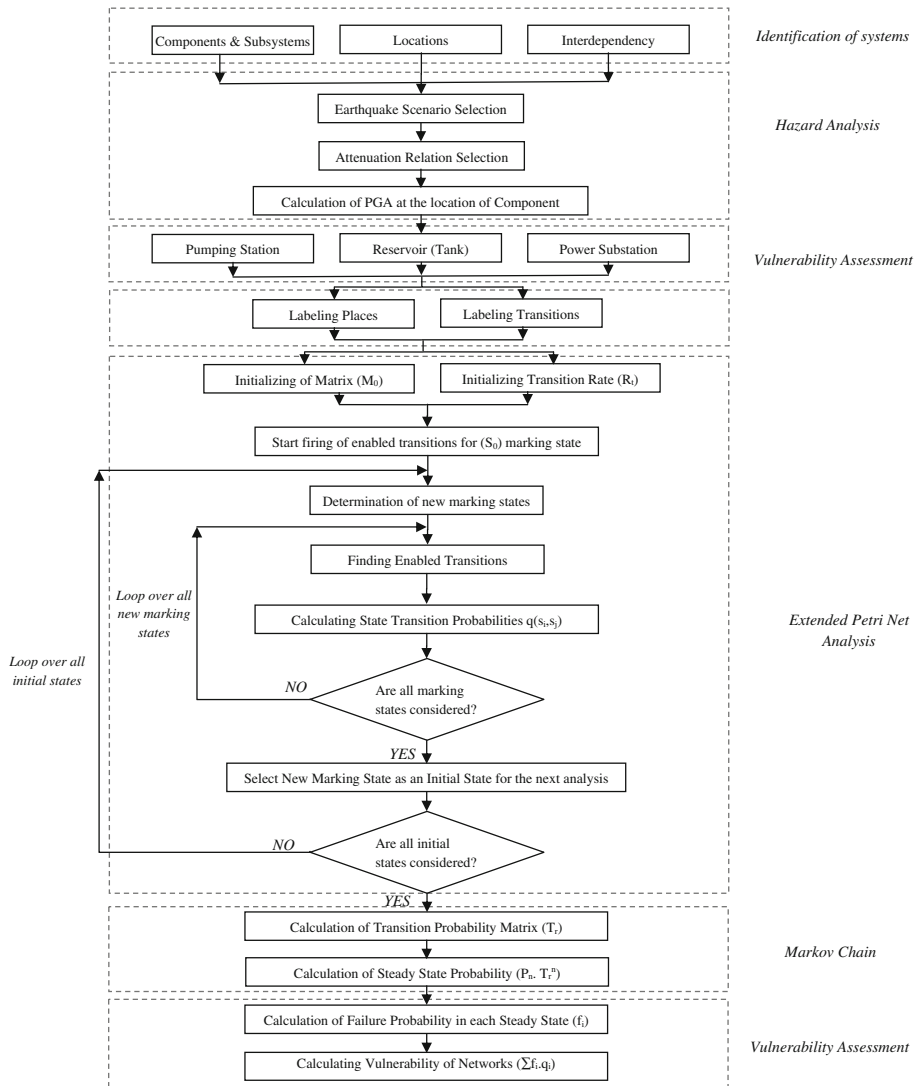


Fig. 1 Proposed algorithm for quantitative vulnerability assessment

2.2 Hazard analysis

The deterministic approach is used in this study. Hazard analysis consists of three steps of selection of earthquake scenario/scenarios, selection of proper attenuation relation/relations and calculation of peak ground acceleration (PGA) at the location of all the components/subsystems.

The vulnerability of considered subsystems, i.e., pumping stations, tanks and power substations, is described using available fragility curves (FEMA 2005).

2.3 Drawing Petri net of systems

In this step, Petri net of the considered network systems is established considering inter-dependency and intra-dependency.

A Petri net is a particular kind of bipartite directed graph, populated by three types of objects. These objects are *places*, *transitions* and *directed arcs*. Directed arcs connect places to transitions or transitions to places. In its simplest form, a Petri net can be represented by a transition together with an input place and an output place. This elementary net may be used to represent various aspects of the modeled systems.

2.4 Vulnerability assessment

A Petri net is formally defined as a 5-tuple $N = (P, T, I, O, M_0)$, where

- (1) $P = \{p_1, p_2, \dots, p_m\}$ is a finite set of places;
- (2) $T = \{t_1, t_2, \dots, t_n\}$ is a finite set of transitions;
- (3) $I: P \times T \rightarrow N$ is an *input function* that defines directed arcs from places to transitions, where N is a set of non-negative integers;
- (4) $O: T \times P \rightarrow N$ is an *output function* that defines directed arcs from transitions to places; and
- (5) $M_0: P \rightarrow N$ is the *initial marking which describes the initial condition of places*.

Directed arcs (arrows) connect places and transitions, with some arcs directed from places to transitions and other arcs directed from transitions to places. An arc directed from a place p_j to a transition t_i defines p_j to be an *input place* of transition t_i , denoted by $I(t_i, p_j) = 1$. An arc directed from a transition t_i to a place p_j defines p_j to be an *output place* of t_i , denoted by $O(t_i, p_j) = 1$. If $I(t_i, p_j) = k$ (or $O(t_i, p_j) = k$), then there exists k directed (parallel) arcs connecting place p_j to transition t_i (or connecting transition t_i to place p_j) (Peterson 1981).

2.5 Extended Petri net analysis

In this step, new states, their relevant transitions and state transition probabilities are determined. Once a transition is enabled, firing is initiated by removing tokens from its input places; throughout the duration of firing, tokens remain in this transition. When firing is terminated, the tokens are distributed in the output places of the transition. If a transition is enabled while it fires, a new independent firing can be initiated.

If state S_j is directly reachable from the state S_i within transition t_k , the transition probability is calculated as

$$q(S_i, S_j) = \frac{r(t_k) \cdot n_i(t_k)}{\sum_{t \in T_i} r(t) \cdot n_i(t)} \quad (1)$$

where T_i is the set of all transitions that originate from state i to other reachable states and $r(t)$ is a firing rate function, which assigns firing rate r to each transition “ t ” in the net, and firing time of transition “ t ” is a random variable $X(t)$ with the distribution function

$$P(X(t) > y) = e^{-r(t)y}, \quad y > 0 \quad (2)$$

$n_i(t)$ is a firing rank function indicating the number of active firings for transition t , i.e., the numbers of firings that have been initiated are not terminated yet.

2.6 Markov chain

The Markov chain modeling approach can simulate the long-term trend of an event. Related theories have been proposed by several researchers in different years (Howard 1971; Kennedy et al. 1974; Karr 1990; Grimstead and Snell 1997; Ching and Ng 2006).

The Markov chain is developed with transition probability matrix T_r whose entries are positive numbers $q(S_i, S_j)$ as positive numbers obtained. A series of consecutive state probabilistic vectors P_0, P_1, P_2 , etc, are obtained from the T_r matrix.

$$P_1 = P_0 \cdot T_r, P_2 = P_1 \cdot T_r, \dots, P_n = P_0 \cdot T_r^n \quad (3)$$

If the state vectors P_n , with increase in n , advance to a constant vector P , the probability vector P is called a steady-state vector for the Markov chain. The entries of P are the long-term probabilities of the Markov chain.

Markov chain can include transient state and absorbing states. Absorbing states have no output and are absorbed within them. This Markov chain can be partitioned simply into transient and absorbing states according to the following canonical form:

$$T_r = \begin{array}{ccc} & Q & \vdots & R \\ & \dots & \vdots & \dots \\ & 0 & \vdots & I \end{array}$$

- I Absorbing state matrix
- 0 Zero matrix
- Q Transient matrix
- R Remaining matrix

The fundamental matrix is

$$N = (I - Q)^{-1} = [n_{ij}] \quad (4)$$

n_{ij} is the predicted number of steps for the case beginning from state i and being absorbed in state j , in transient states.

t_i as the expected number of steps before absorption of the chain, with the assumption of beginning from state i , can be calculated as

$$t_i = N \times C = \sum_{j=1}^n n_{ij} \quad (5)$$

C Vertical vector with all entries set to 1.

If b_{ij} is the absorption probability in absorbing state S_j that has begun from the transient state S_i , and B is a matrix with b_{ij} entries, then

$$B = N \times R. \quad (6)$$

2.7 Vulnerability assessment of networks

The absorbing states and the set of transitions for reaching each absorbing state (known as track) are calculated in the previous steps. For each final absorbing state, the firing

probability of all transitions and the failure probability of all related components/subsystems are multiplied to calculate the absolute probability of the considered failure state. By summing the failure probabilities of all calculated absorbing states, the vulnerability of the assumed network is achieved.

3 Case study

In this paper, a part of the power and water distribution networks of one of the municipal districts of Tehran is investigated as a case study in order to realize their interactive effects. This district covers approximately 64 square kilometers, comprising 9 % of Tehran's area. It has around 650,000 inhabitants (8.2 % of Tehran's population). The selected district is chosen because of the availability of a considerable number of different vulnerable components, i.e., power substations, storage tanks and pumping stations. Another reason for this selection is that most of the structures in the district have residential occupancy resulting in high water demand in case of earthquake occurrence. The North Tehran fault also passes through the northern parts of the district that indicates its high seismicity. So it is necessary to investigate the seismic performance of infrastructures of the district.

As depicted in Fig. 2, the only 230/63 kv substation in this area is power substation 1, which is an ASEA type substation and feeds three 63/20 kv substations, power substation 2, power substation 3 and power substation 4. There are four water storage tanks in this area. Except for storage tank 4, all the other tanks have a pumping system that conveys the water to higher storage tanks.

Among the active faults in this area, the most threatening ones are the Mosha fault, North Tehran fault and South Rey fault. With regard to the earthquake scenario, the most devastating earthquakes should be taken into consideration. According to the investigations of Japan International Cooperation Agency, four models are proposed for the earthquake scenario: the Mosha fault model, the North Tehran fault model, the Rey fault model and the floating model (JICA 2006). The floating model is selected in this study. According to this scenario, the PGA in the area is equal to 0.35 g.

3.1 Network model

The Petri net modeling and the Markov chain analysis are used in order to evaluate the failure risk of a network of power and water infrastructures in the district. Initially, the models should be created so that their interactions can be depicted both graphically and in matrices. The network should then be analyzed using extended Petri net and Markov chain methods. Because the main purpose of this article is to study the level of interaction and influence of the power network on the water network, both types of network should be considered:

1. Water network model (disregarding its interaction with the power network)
2. Power and water network models

After preparing these models, considering the intensity of the scenario earthquake and its return period, the models are analyzed considering all possible event combinations and their occurrence probability. The probabilities of different combinations of transitions that lead to the failure of all facilities are calculated with the Markov chain. Summing these probabilities in each model, the vulnerability of the each system against earthquake is

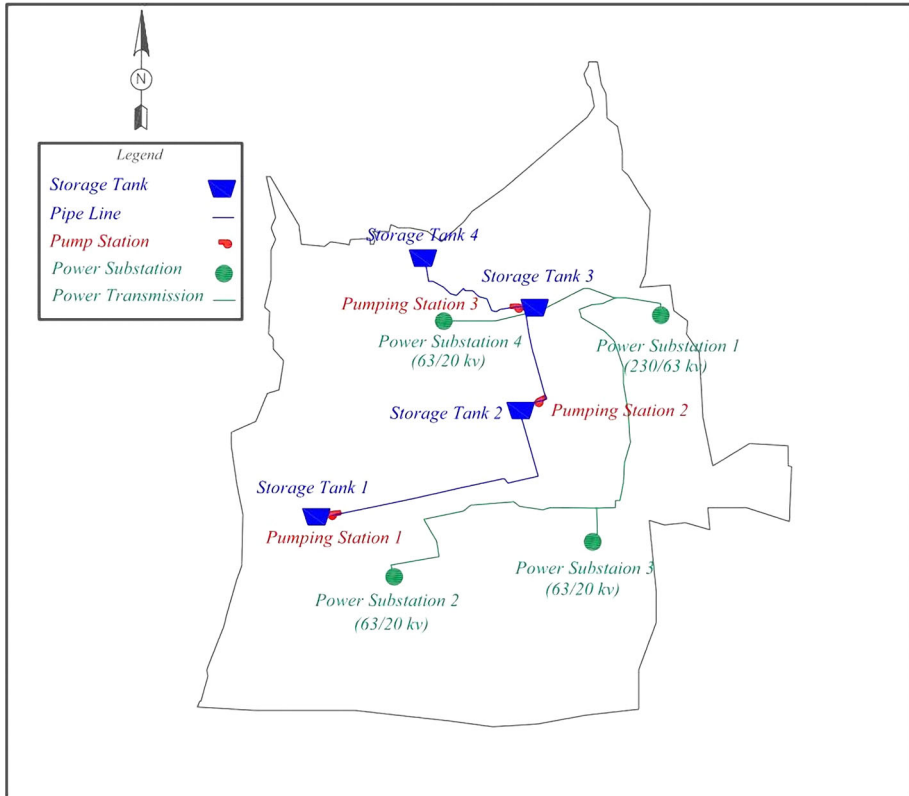


Fig. 2 Power and water distribution networks

obtained. Finally, their vulnerabilities are compared and the influence of the power network on the water network is discovered.

In the present research, only the important components of the networks are considered. Hence, the components of the water network include the water storage tanks, pumping stations and the components of the power network, which include the 230/63 kv and 63/20 kv substations for the area.

Having the amount of PGA in this area (0.35 g), the values of failure probability for each component are acquired from the corresponding fragility curves (FEMA 2005). In the absence of local fragility curves for infrastructure components, relevant FEMA's fragility curves are used in this study. Table 1 shows the network components and their failure probability.

3.2 Modeling the networks with the Petri net

In order to model the networks with the Petri net, the location and connections of the components and their relations should be clarified according to the configuration plan. The corresponding substation supply power for each pumping station according to Fig. 2 is shown in Table 2.

Table 1 Network components and their failure probability (FEMA 2005)

Network components	Damage states	Failure probability
230/63 kv substation with standard components	Moderate	0.87
63/20 kv substation with standard components	Moderate	0.74
Unanchored small pumping station	Moderate	0.67
Unanchored concrete tank	Moderate	0.40

By determination of the facilities' dependencies, the list of places and transitions of water network model and power network model and the combined model could be provided. It should be noted that the considered damage rate of places in the models (network facilities) is selected based on their failure probabilities, i.e., it is considered that firing rates are proportional to the failure probabilities, which are derived from their fragility curves. Moreover, the cases of transitions with a firing rate equal to 1 indicate that following a failure in the previous place, the next place will stop as well.

3.2.1 The Petri net of water network

In the water network model, four water storage tanks and three pumping stations are considered. In this model, 22 places (see Table 3) and 14 transitions (see Table 4) are defined. Given that some of the places in the network have multiple inputs from different transitions, in order to avoid multiple marking of one place by different input transitions, inhibitor arcs were utilized. Put simply, if a water storage tank fails because of the earthquake, the disruption caused by the probable destruction of its feeding tank on the tank will not be considered. The initial marking of the water network is

$$M_0 = 1000000001111111111111$$

This indicates that before the occurrence of the earthquake, all components are working.

In this model, when the transition 1 fires, the earthquake occurs. Hence, it is possible that transitions 2–8 fire afterward, leading to failure in water storage tanks or pumping stations. This procedure continues until the entire network fails (all main places become marked). The Petri net model of the water network is shown in Fig. 3.

3.2.2 The Petri net model of power network

In the power network model, four water storage tanks and three pumping stations are considered. In this model, 13 places (see Table 5) and 8 transitions (see Table 6) are defined. The initial marking of power network is

$$M_0 = 1000001111111$$

This indicates that before the occurrence of the earthquake, all power components are working.

In this model, when the transition 1 fires, the earthquake occurs. Hence, it is possible that transitions 2, 4, 6 and 7 fire afterward, leading to failure in power substations. This procedure continues until the entire network fails (all main places become marked). The Petri net model of the power network is shown in Fig. 4.

Table 2 Power substations that supply pumping stations

Pumping station	(Pumping station 1)	(Pumping station 2)	(Pumping station 3)
Power substation	(Power substation 2)	(Power substation 3)	(Power substation 4)

Table 3 List of places in the water network

Places (p_i)	Description
p_1	Nothing happened
p_2	Occurrence of earthquake
p_3	Failure in “storage tank 2”
p_4	Failure in “pumping station 2”
p_5	Failure in “pumping station 1”
p_6	Failure in “storage tank 1”
p_7	Failure in “pumping station 3”
p_8	Failure in “storage tank 4”
p_9	Failure in “storage tank 3”
p_{10}	“Occurrence of earthquake” (dummy place for “pumping station 2”)
p_{11}	“Occurrence of earthquake” (dummy place for “pumping station 1”)
p_{12}	“Occurrence of earthquake” (dummy place for “storage tank 1”)
p_{13}	Failure in “storage tank 1” (dummy place for “storage tank 2”)
p_{14}	Failure in “pumping station 1” (virtual place for “storage tank 2”)
p_{15}	Failure in “pumping station 2” (dummy place for “storage tank 3”)
p_{16}	“Occurrence of earthquake” (dummy place for “storage tank 2”)
p_{17}	“Occurrence of earthquake” (dummy place for “pumping station 3”)
p_{18}	“Occurrence of earthquake” (dummy place for “storage tank 4”)
p_{19}	“Occurrence of earthquake” (dummy place for “storage tank 3”)
p_{20}	Failure in “storage tank 2” (dummy place for “storage tank 3”)
p_{21}	Failure in “pumping station 3” (dummy place for “storage tank 4”)
p_{22}	Failure in “storage tank 3” (virtual place for “storage tank 4”)

3.2.3 Petri net model of combined power and water networks

In this section, a combined model is proposed that considers the relations of power and water networks. In this network, the power supply of pumping station connects the 2 networks, so that failure in a power station results in the failure of the correspondent pumping station. Thus, a combined model of water and power networks is provided, as shown in Fig. 5. Power substations, water tanks and pumping stations have been indicated, respectively, in green, blue and red. List of combined network places and transitions is included in Tables 7 and 8, respectively.

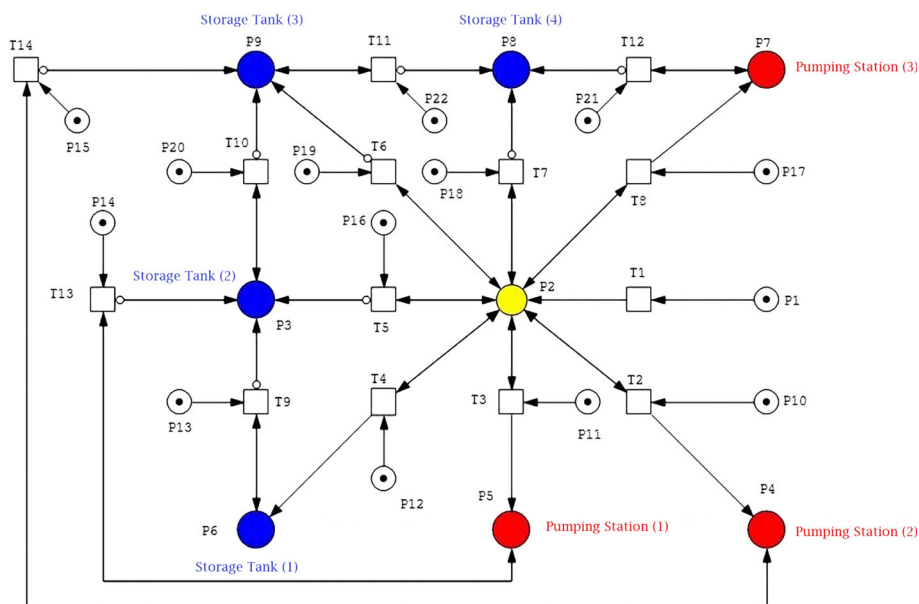
The initial marking of power network is

$$M_0 = 1000000001111111111111100001111111111$$

Initially, nothing has happened and transition t_1 is ready to fire. The network begins with the firing of t_1 , or the occurrence of the earthquake, and the marker positioned in place one of marking vector (P_1) moves to place two (P_2), indicating that the earthquake occurred.

Table 4 List of transitions in the water network

Transition (t_i)	Description	Firing rate (r_i)
t_1	The earthquake occurs	Deterministic
t_2	“Occurrence of earthquake” affects “pumping station 2”	0.67
t_3	“Occurrence of earthquake” affects “pumping station 1”	0.67
t_4	“Occurrence of earthquake” affects “storage tank 1”	0.4
t_5	“Occurrence of earthquake” affects “storage tank 2”	0.4
t_6	“Occurrence of earthquake” affects “storage tank 3”	0.4
t_7	“Occurrence of earthquake” affects “storage tank 4”	0.4
t_8	“Occurrence of earthquake” affects “pumping station 3”	0.67
t_9	Failure in “storage tank 1” affects “storage tank 2”	1
t_{10}	Failure in “storage tank 2” affects “storage tank 3”	1
t_{11}	Failure in “storage tank 3” affects “storage tank 4”	1
t_{12}	Failure in “pumping station 3” affects “storage tank 4”	1
t_{13}	Failure in “pumping station 1” affects “storage tank 2”	1
t_{14}	Failure in “pumping station 2” affects “storage tank 3”	1

**Fig. 3** Petri net model of water network

Hence, the first state of failure appears (S_1). In this state, P_2 is marked and other transitions are ready to fire. Upon the firing of one, the present markers move to the next places and the consequent states of failure are defined. This procedure continues until all main places become marked. Marking these places indicates that all the facilities have stopped working and the system has totally failed.

Table 5 List of power network places

Places (p_i)	Description
p_1	Nothing happened
p_2	“Occurrence of earthquake”
p_3	Failure in “power substation 1”
p_4	Failure in “power substation 2”
p_5	Failure in “power substation 3”
p_6	Failure in “power substation 4”
p_7	“Occurrence of earthquake” (dummy place for “power substation 1”)
p_8	Failure in “power substation 1” (dummy place for “power substation 2”)
p_9	“Occurrence of earthquake” (dummy place for “power substation 2”)
p_{10}	Failure in “power substation 1” (dummy place for “power substation 3”)
p_{11}	“Occurrence of earthquake” (dummy place for “power substation 3”)
p_{12}	“Occurrence of earthquake” (dummy place for “power substation 4”)
p_{13}	Failure in “power substation 1” (dummy place for “power substation 4”)

Table 6 List of power network transitions

Transition (t_i)	Description	Firing rate (r_i)
t_1	The earthquake occurs	Deterministic
t_2	“Occurrence of earthquake” affects “power substation 1”	0.87
t_3	Failure in “power substation 1” affects “power substation 2”	1
t_4	“Occurrence of earthquake” affects “power substation 2”	0.74
t_5	Failure in “power substation 1” affects “power substation 3”	1
t_6	“Occurrence of earthquake” affects “power substation 3”	0.74
t_7	“Occurrence of earthquake” affects “power substation 4”	0.74
t_8	Failure in “power substation 1” affects “power substation 4”	1

The initial state (S_i) and the next states (S_j) in order of events, order of fired transitions in previous states (*Track*), current transitions (t) and their rates [$r(t)$], and transition probability (q) can be presented in a table. Additionally, the infrastructures that fail upon firing of their correspondent transition can be included in this table. The occurrence probability of ready transitions in each state is derived from Eq (1). Considering the number of places and transitions of the models of these two networks, the number of calculated states will be large and impossible to provide in one table. Tables 9 and 10 show some states regarding the water network. The calculations show that there are 458 states for the water network, whereas this number is 43,508 for the power and water networks.

3.3 Markov chain analysis of the networks

After considering all failure states (S) in the model and the calculation of the transition probability from one state to another (q) based on its firing rate (r), the transition matrix (T_r) is formed. T_r is a square matrix with the number of columns and rows equal to the number of states derived from the model. The entries of this matrix are the values of transition probability (q) from one state to another.

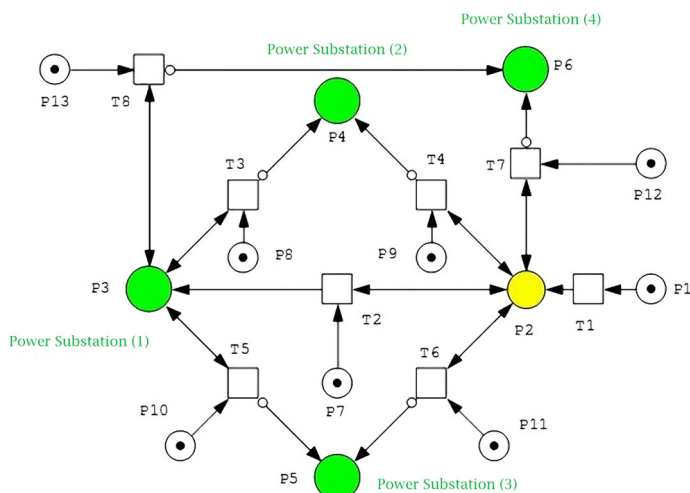


Fig. 4 Petri net model of power network

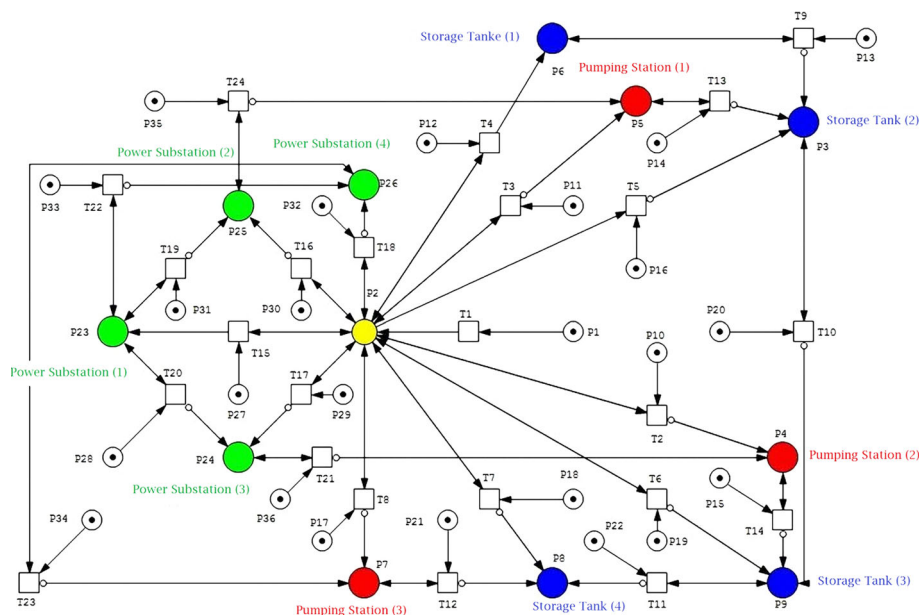


Fig. 5 Petri net model of water and power networks

The transition matrix of water network is 485×485 , and the transition matrix of power and water networks is $43,508 \times 43,508$.

In order to calculate the values of steady-state probability, the transition matrix should be multiplied to the initial state matrix. In steady states, all elements of the network are out of order. It is worth mentioning that at the first glance, finding the steady-state solution of such an enormous Markov chain might seem infeasible. However, by taking a look at the

Table 7 List of combined network places

Places (p_i)	Description
p_1	Nothing happened
p_2	“Occurrence of earthquake”
p_3	Failure in “storage tank 2”
p_4	Failure in “pumping station 2”
p_5	Failure in “pumping station 1”
p_6	Failure in “storage tank 1”
p_7	Failure in “pumping station 3”
p_8	Failure in “storage tank 4”
p_9	Failure in “storage tank 3”
p_{10}	“Occurrence of earthquake” (dummy place for “pumping station 2”)
p_{11}	“Occurrence of earthquake” (dummy place for “pumping station 1”)
p_{12}	“Occurrence of earthquake” (dummy place for “storage tank 1”)
p_{13}	Failure in “storage tank 1” (dummy place for “storage tank 2”)
p_{14}	Failure in “pumping station 1” (dummy place for “storage tank 2”)
p_{15}	Failure in “pumping station 2” (dummy place for “storage tank 3”)
p_{16}	“Occurrence of earthquake” (dummy place for “storage tank 2”)
p_{17}	“Occurrence of earthquake” (dummy place for “pumping station 3”)
p_{18}	“Occurrence of earthquake” (dummy place for “storage tank 4”)
p_{19}	“Occurrence of earthquake” (dummy place for “storage tank 3”)
p_{20}	Failure in “storage tank 2” (dummy place for “storage tank 3”)
p_{21}	Failure in “pumping station 3” (dummy place for “storage tank 4”)
p_{22}	Failure in “storage tank 3” (dummy place for “storage tank 4”)
p_{23}	Failure in “power substation 1”
p_{24}	Failure in “power substation 3”
p_{25}	Failure in “power substation 2”
p_{26}	Failure in “power substation 4”
p_{27}	“Occurrence of earthquake” (dummy place for “power substation 1”)
p_{28}	Failure in “power substation 1” (dummy place for “power substation 2”)
p_{29}	“Occurrence of earthquake” (dummy place for “power substation 2”)
p_{30}	Failure in “power substation 1” (dummy place for “power substation 3”)
p_{31}	“Occurrence of earthquake” (dummy place for “power substation 3”)
p_{32}	“Occurrence of earthquake” (dummy place for “power substation 4”)
p_{33}	Failure in “power substation 1” (dummy place for “power substation 4”)
p_{34}	Failure in “power substation 4” (dummy place for “pumping station 3”)
p_{35}	Failure in “power substation 2” (dummy place for “pumping station 1”)
p_{36}	Failure in “power substation 3” (dummy place for “pumping station 2”)

gigantic transition matrix and considering the structure of the problem, one will realize that at each row of this matrix, most elements are zero. In other words, the transition matrix of the problem is sparse. As a result, software was developed by authors to exploit this characteristic feature in the calculations. In practice, the program which was written in C++ only took a few minutes to solve the problem on an ordinary machine at the time with two gigabytes of memory and under Linux operating system.

Transition (t_i)	Description	Transition rate (r_i)
t_1	The earthquake occurs	Deterministic
t_2	“Occurrence of earthquake” affects “pumping station 2”	0.67
t_3	“Occurrence of earthquake” affects “pumping station 1”	0.67
t_4	“Occurrence of earthquake” affects “storage tank 1”	0.4
t_5	“Occurrence of earthquake” affects “storage tank 2”	0.4
t_6	“Occurrence of earthquake” affects “storage tank 3”	0.4
t_7	“Occurrence of earthquake” affects “storage tank 4”	0.4
t_8	“Occurrence of earthquake” affects “pumping station 3”	0.67
t_9	Failure in “storage tank 1” affects “storage tank 2”	1
t_{10}	Failure in “storage tank 2” affects “storage tank 3”	1
t_{11}	Failure in “storage tank 3” affects “storage tank 4”	1
t_{12}	Failure in “pumping station 3” affects “storage tank 4”	1
t_{13}	Failure in “pumping station 1” affects “storage tank 2”	1
t_{14}	Failure in “pumping station 2” affects “storage tank 3”	1
t_{15}	“Occurrence of earthquake” affects the “power substation 1”	0.87
t_{16}	Failure in “power substation 1” affects “power substation 2”	0.74
t_{17}	The earthquake affects “power substation 2”	0.74
t_{18}	Failure in “power substation 1” affects “power substation 3”	0.74
t_{19}	“Occurrence of earthquake” affects “power substation 3”	1
t_{20}	“Occurrence of earthquake” affects “power substation 4”	1
t_{21}	Failure in “power substation 1” affects “power substation 4”	1
t_{22}	Failure in “power substation 4” affects “pumping station 3”	1
t_{23}	Failure in “power substation 2” affects “pumping station 1”	1
t_{24}	Failure in “power substation 3” affects “pumping station 2”	1

Table 9 Some initial states in the water network model

S_i	Track	t_k	$r(t)$	S_j	Damaged infrastructure place	Transition probability q
S_0	–	1	Deterministic	S_1	p_2	1
S_1	1	2	0.67	S_2	p_2, p_4	0.185596
	1	3	0.67	S_3	p_2, p_5	0.185596
	1	4	0.4	S_4	p_2, p_6	0.110803
	1	5	0.4	S_5	p_2, p_3	0.110803
	1	6	0.4	S_6	p_2, p_9	0.110803
	1	7	0.4	S_7	p_2, p_8	0.110803
	1	8	0.67	S_8	p_2, p_7	0.185596
S_2	1,2	3	0.67	S_9	p_2, p_4, p_5	0.170051
	1,2	4	0.4	S_{10}	p_2, p_4, p_6	0.101523
	1,2	5	0.4	S_{11}	p_2, p_4, p_3	0.101523
	1,2	6	0.4	S_{12}	p_2, p_4, p_9	0.101523
	1,2	7	0.4	S_{13}	p_2, p_4, p_8	0.101523
	1,2	8	0.67	S_{14}	p_2, p_4, p_7	0.170051
	1,2	14	0.4	S_{15}	p_2, p_4, p_9	0.253807
S_3	1,3	2	0.67	S_9	p_2, p_5, p_4	0.170051
	1,3	4	0.4	S_{16}	p_2, p_5, p_6	0.101523
	1,3	5	0.4	S_{17}	p_2, p_5, p_3	0.101523
	1,3	6	0.4	S_{18}	p_2, p_5, p_9	0.101523
	1,3	7	0.4	S_{19}	p_2, p_5, p_8	0.101523
	1,3	8	0.67	S_{20}	p_2, p_5, p_7	0.170051
	1,3	13	1	S_{21}	p_2, p_5, p_3	0.253807

$$P * T_r^8 = [0.00000000 \dots 0.0176 \ 0.0179 \ 0.0184 \ 0.028 \ 0.0282 \ 0.022 \ 0.029]$$

This procedure was also taken for the power and water networks, and 1,728 steady states were obtained for this model. A part of the probability vector for steady states is shown below:

$$P * T_r^{13} = [0.00000000 \dots 6.46672e - 05 \ 8.8354e - 05 \ 6.2589e - 05 \ 3.62439e - 05]$$

3.4 Calculating the vulnerability (The probability of failure)

In this analysis, the vulnerability of the network is considered in a damage state that all the network facilities are failed. Two parameters should be taken into account for the calculation:

1. q_i : Steady-state probability in each network
2. f_i : Values of failure probability in each steady state

The first parameter q_i was obtained in the previous section. In order to calculate the second parameter f_i , after identification of fired transitions in each steady state, the failure probability of all of them (firing probabilities of the correspondent transitions) should be multiplied together.

Table 10 Some final states in the water network model

S_i	Track	t_k	$r(t)$	S_j	Damaged infrastructure	q
S_{436}	1, 4, 9, 10, 11, 8, 2	3	0.67	S_{477}	$p_2, p_6, p_3, p_9, p_8, p_7, p_4, p_5$	1
S_{437}	1, 8, 12, 4, 9, 10, 2	3	0.67	S_{478}	$p_2, p_7, p_8, p_6, p_3, p_9, p_4, p_5$	1
S_{438}	1, 2, 14, 8, 12, 4, 9	3	0.67	S_{479}	$p_2, p_4, p_9, p_7, p_8, p_6, p_3, p_5$	1
S_{439}	1, 2, 14, 11, 4, 9, 8	3	0.67	S_{480}	$p_2, p_4, p_9, p_8, p_6, p_3, p_7, p_5$	1
S_{440}	1, 8, 7, 6, 5, 4, 3	2	0.67	S_{458}	$p_2, p_7, p_8, p_9, p_3, p_6, p_5, p_4$	1
S_{441}	1, 6, 11, 8, 5, 4, 3	2	0.67	S_{459}	$p_2, p_9, p_8, p_7, p_3, p_6, p_5, p_4$	1
S_{442}	1, 8, 12, 6, 5, 4, 3	2	0.67	S_{460}	$p_2, p_7, p_8, p_9, p_3, p_6, p_5, p_4$	1
S_{443}	1, 5, 10, 8, 7, 4, 3	2	0.67	S_{461}	$p_2, p_3, p_9, p_7, p_8, p_6, p_5, p_4$	1
S_{444}	1, 5, 10, 11, 8, 4, 3	2	0.67	S_{463}	$p_2, p_3, p_9, p_8, p_7, p_6, p_5, p_4$	1
S_{445}	1, 8, 12, 5, 10, 4, 3	2	0.67	S_{464}	$p_2, p_7, p_8, p_3, p_9, p_6, p_5, p_4$	1
S_{446}	1, 4, 9, 8, 7, 6, 3	2	0.67	S_{467}	$p_2, p_6, p_3, p_7, p_8, p_4, p_5, p_4$	1
S_{447}	1, 3, 13, 8, 7, 6, 4	2	0.67	S_{468}	$p_2, p_5, p_3, p_7, p_8, p_9, p_6, p_4$	1
S_{448}	1, 6, 11, 4, 9, 8, 3	2	0.67	S_{469}	$p_2, p_9, p_8, p_6, p_3, p_7, p_5, p_4$	1
S_{449}	1, 8, 12, 4, 9, 6, 3	2	0.67	S_{470}	$p_2, p_7, p_8, p_6, p_3, p_9, p_5, p_4$	1
S_{450}	1, 3, 13, 6, 11, 8, 4	2	0.67	S_{471}	$p_2, p_5, p_3, p_9, p_8, p_7, p_6, p_4$	1
S_{451}	1, 3, 13, 8, 12, 6, 4	2	0.67	S_{472}	$p_2, p_5, p_3, p_7, p_8, p_9, p_6, p_4$	1
S_{452}	1, 4, 9, 10, 8, 7, 3	2	0.67	S_{473}	$p_2, p_6, p_3, p_9, p_7, p_8, p_5, p_4$	1
S_{453}	1, 3, 13, 10, 8, 7, 4	2	0.67	S_{475}	$p_2, p_5, p_3, p_9, p_7, p_7, p_6, p_4$	1
S_{454}	1, 4, 9, 10, 11, 8, 3	2	0.67	S_{477}	$p_2, p_6, p_3, p_9, p_8, p_7, p_5, p_4$	1
S_{455}	1, 8, 12, 4, 9, 10, 3	2	0.67	S_{478}	$p_2, p_7, p_8, p_6, p_3, p_9, p_5, p_4$	1
S_{456}	1, 3, 13, 10, 8, 12, 4	2	0.67	S_{481}	$p_2, p_5, p_3, p_9, p_7, p_8, p_6, p_4$	1
S_{457}	1, 3, 13, 10, 11, 8, 4	2	0.67	S_{483}	$p_2, p_5, p_3, p_9, p_8, p_7, p_6, p_4$	1

Now, if the relative probability of each steady state (the relative probability of that steady-state scenario with respect to all the steady-state scenarios) is multiplied to the failure probability of the same state and the final results are summed, the obtained value shows the vulnerability of the modeled network against the considered earthquake.

Table 11 shows the vulnerability of the water network.

The vulnerability assessment has been performed for the studied networks in 8 different conditions. The calculated conditions can be classified into 4 groups:

1. Components are independent in one network.
2. Components are intra-dependent in one network.
3. Networks are independent.
4. Networks are interdependent.

Failure probability of all facilities has been calculated in different conditions, as shown in Table 12.

With the assumption of independency of power and water networks, in order to assess the probability of simultaneous failure in both networks (so that facilities of both networks fail), their failure probabilities of conditions 2 and 4 should be multiplied. The final failure probability for water and power networks is 0.024434.

In another condition, it is assumed that the power network is damaged by the earthquake and has stopped working; it is possible to analyze the conditional failure of the water network considering the effect of the failure in the power network on the water network,

Table 11 Some final states in the water network model

S_i	Track	f_i	q_i	$f_i \cdot q_i$
S_{458}	1, 8, 7, 6, 5, 4, 3, 2	0.00769953	0.0658785	0.000507233
S_{459}	1, 6, 11, 8, 5, 4, 3, 2	0.0192488	0.0693628	0.001335151
S_{460}	1, 8, 12, 6, 5, 4, 3, 2	0.0192488	0.0419863	0.000808186
S_{461}	1, 5, 10, 8, 7, 4, 3, 2	0.0192488	0.0685388	0.00131929
S_{462}	1, 2, 14, 8, 7, 5, 4, 3	0.0192488	0.0480399	0.00092471
S_{463}	1, 5, 10, 11, 8, 4, 3, 2	0.0481221	0.0497883	0.002395918
S_{464}	1, 8, 12, 5, 10, 4, 3, 2	0.0481221	0.0460958	0.002218227
S_{465}	1, 2, 14, 8, 12, 5, 4, 3	0.0481221	0.0322100	0.001550013
S_{466}	1, 2, 14, 11, 8, 5, 4, 3	0.0481221	0.0356346	0.001714812
S_{467}	1, 4, 9, 8, 7, 6, 3, 2	0.0192488	0.0345201	0.000664471
S_{468}	1, 3, 13, 8, 7, 6, 4, 2	0.0192488	0.0543662	0.001046484
S_{469}	1, 6, 11, 4, 9, 8, 3, 2	0.0481221	0.0342321	0.001647321
S_{470}	1, 8, 12, 4, 9, 6, 3, 2	0.0481221	0.0222494	0.001070688
S_{471}	1, 3, 13, 6, 11, 8, 4, 2	0.0481221	0.0539759	0.002597434
S_{472}	1, 3, 13, 8, 12, 6, 4, 2	0.0481221	0.0350335	0.001685886
S_{473}	1, 4, 9, 10, 8, 7, 3, 2	0.0481221	0.0252735	0.001216214
S_{474}	1, 2, 14, 4, 9, 8, 7, 3	0.0481221	0.0264340	0.00127206
S_{475}	1, 3, 13, 10, 8, 7, 4, 2	0.0481221	0.0401080	0.001930081
S_{476}	1, 2, 14, 3, 13, 8, 7, 4	0.0481221	0.0415938	0.002001581
S_{477}	1, 4, 9, 10, 11, 8, 3, 2	0.120305	0.0137338	0.001652245
S_{478}	1, 8, 12, 4, 9, 10, 3, 2	0.120305	0.017633	0.002121338
S_{479}	1, 2, 14, 8, 12, 4, 9, 3	0.120305	0.0178986	0.002153291
S_{480}	1, 2, 14, 11, 4, 9, 8, 3	0.120305	0.0183786	0.002211037
S_{481}	1, 3, 13, 10, 8, 12, 4, 2	0.120305	0.0279573	0.003363403
S_{483}	1, 2, 14, 3, 13, 8, 12, 4	0.120305	0.0281587	0.003387632
S_{483}	1, 3, 13, 10, 11, 8, 4, 2	0.120305	0.0219654	0.002642547
S_{485}	1, 2, 14, 11, 3, 13, 8, 4	0.120305	0.028953	0.003483191
$\sum q_i = 1$			$\sum f_i \cdot q_i = 0.0511284$	

without assessing the vulnerability of the power network. In order to consider this condition, the failure probability of power station transitions should be set equal to 1 in both the model and the matrices. Through making these changes and the calculations, the results are obtained. This method is in fact a systematic removal.

4 Results and discussion

1. In water network, considering the intra-dependencies of components, the failure probability increased 6.6-fold (0.051128/0.007699), compared to the case that components work independently.
2. In power network, considering the intra-dependencies of components, the failure probability increased 1.4-fold (0.477910/0.352544), compared to the case that components work independently.

Table 12 Failure probability for water and power networks in different conditions

Condition No.	Network	Description	Failure probability
1	Water	Independent components	0.007699
2	Water	Intra-dependent components	0.051128
3	Power	Independent components	0.352544
4	Power	Intra-dependent components	0.477910
5	Water and power	Independent networks	0.002714
		Independent components	
6	Water and power	Independent networks	0.024434
		Intra-dependent components	
7	Water and power	Interdependent networks	0.0460198
		Inter- and intra-dependent components	
8	Water and power	Interdependent networks	0.084828
		Interdependent components (power network is considered failed initially)	

3. The probability of simultaneous failure of power and water networks was calculated when the components in each network work both intra-dependent and independent. Considering the intra-dependencies of components, the failure probability grows nine fold ($0.024434/0.002714$).
4. The simultaneous failure probability calculated for independent power and water networks with intra-dependent components was 0.024434, whereas this value was obtained 0.0460198 for a combined model of power and water networks with interdependent components. Comparison of these numbers implies that the failure probability of the entire network has almost doubled. Thus, it can be inferred that the interdependency between these networks has increased their vulnerability.
5. With the assumption of total failure in power network, two models were considered:
 - a. Independent water network with intra-dependent components
 - b. Dependent water and power networks with interdependent components

The comparison indicates that failure probability of the water network that is dependent on the failed power network is 1.66 of the independent water network ($0.084828/0.051128$).

Thus, it can be concluded that failure in the power network will increase the vulnerability of water networks.

Concerning the studied networks, certain approaches are proposed in order to decrease the vulnerability:

- Utilization of backup power systems for the facilities of the water network, such as water storage tanks, pumping stations and well facilities.
- Utilization of more than one power substation for several facilities of the water network.
- Utilization of more than one input feeder for 63/20 substations in order to be fed from multiple 230/63 kv substations.
- Ring configuration design for distribution networks, so that in case of damage to one component, the energy can be supplied from other routes.

5 Conclusion

Metropolitan areas consist of complicated systems of interconnected infrastructures that are highly interdependent. The results from analysis of one infrastructure as an independent system are not realistic without considering the behavior of other interconnected infrastructures. Consequently, the study of the interdependencies among critical infrastructures is important for addressing the cascading effects of a failed infrastructure on the entire network to properly model its performance and help the disaster management team in decision making.

For performing a quantitative vulnerability assessment of interdependent infrastructures against earthquake, a case study of power and water distribution networks of one of the municipal districts of Tehran was conducted.

The extended Petri net and Markov chain were used to demonstrate the effect of interdependencies among the above-stated infrastructure systems on the vulnerability assessment results.

In this research, three cases were assessed quantitatively:

- The intra-dependency effects of different components in each network

- The interdependency effects between the considered critical infrastructures
- The behavior of the water network considering intra- and interdependency, when the power network fails.

Failure probabilities of all facilities were calculated in different conditions. The analyses showed that considering the mentioned interdependencies has a major influence on their performance simulations and assessment of their exact vulnerability. For example, the probability of simultaneous failure of power and water networks was calculated when the components in each network work both intra-dependent and independent. Considering the intra-dependencies of components, the failure probability grows ninefold in this case.

With the assumption of total failure in power network, two models were considered:

- c. Independent water network with intra-dependent components
- d. Dependent water and power networks with interdependent components

The comparison indicates that failure probability of water network that is dependent on the failed power network is 1.66 of the independent water network.

Thus, it can be concluded that failure in the power network will increase the vulnerability of water networks.

The analyses showed that considering the mentioned interdependencies has a major influence on the performance simulations and assessment of the exact vulnerability.

It is quantitatively shown that the more interdependent the infrastructures are, the more vulnerable they are. Eventually, these quantitative results could be used in design, restoration and disaster management planning for safety assessment of critical infrastructures.

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